Hands-on Introduction to NGS Data Analysis

Lecturers: Lucrezia Valeriani[1,2], Giorgia Gandolfi[2,3]

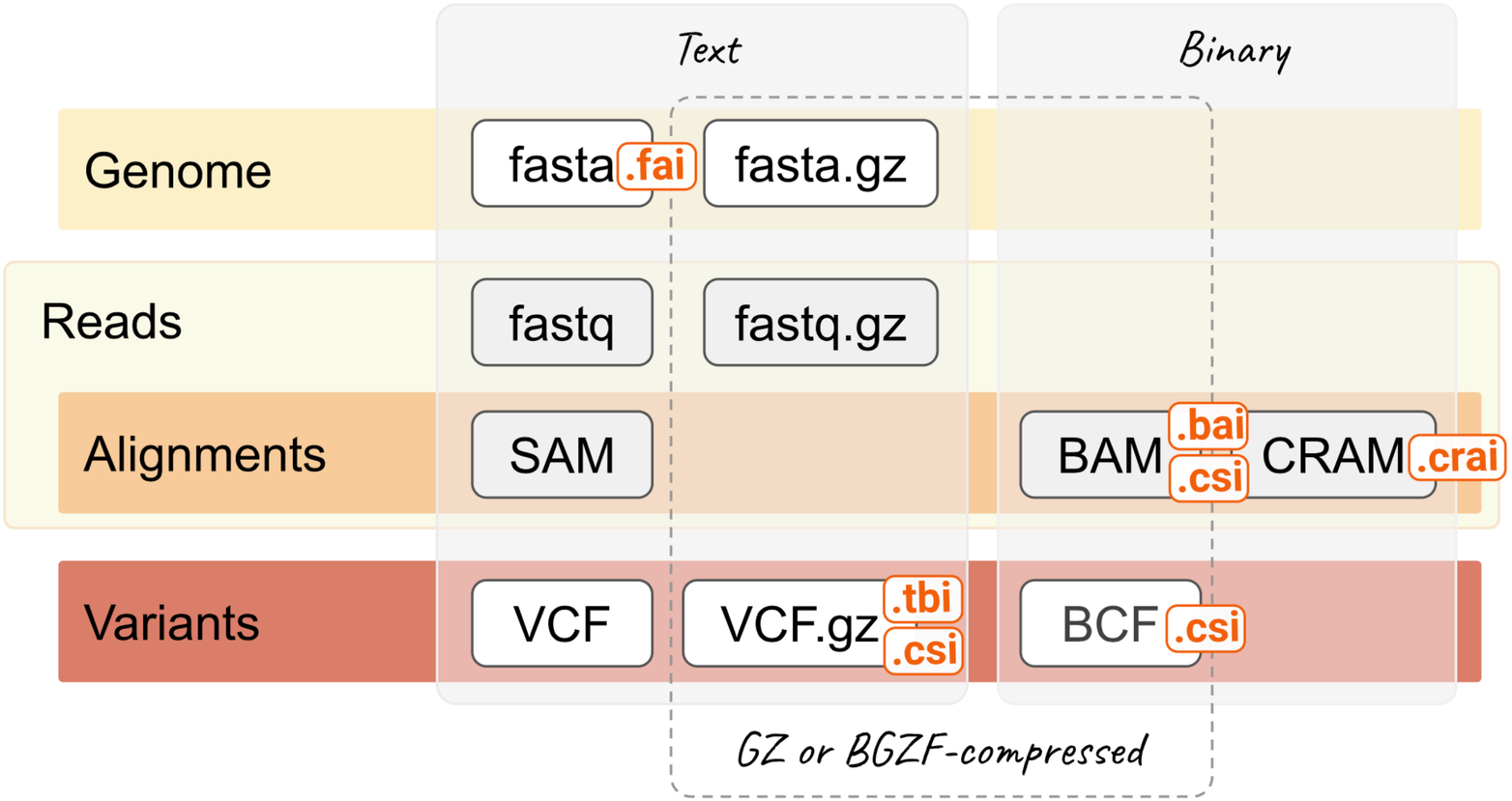
[1]AREA Science Park, [2]University of Trieste, [3]IRCCS Ospedale San Raffaele

Organisers:
Giulio Caravagna, PI of CDSLab - University of Trieste
Stefano Cozzini, RIT - AREA Science Park

Day 2 - 12 November 2026



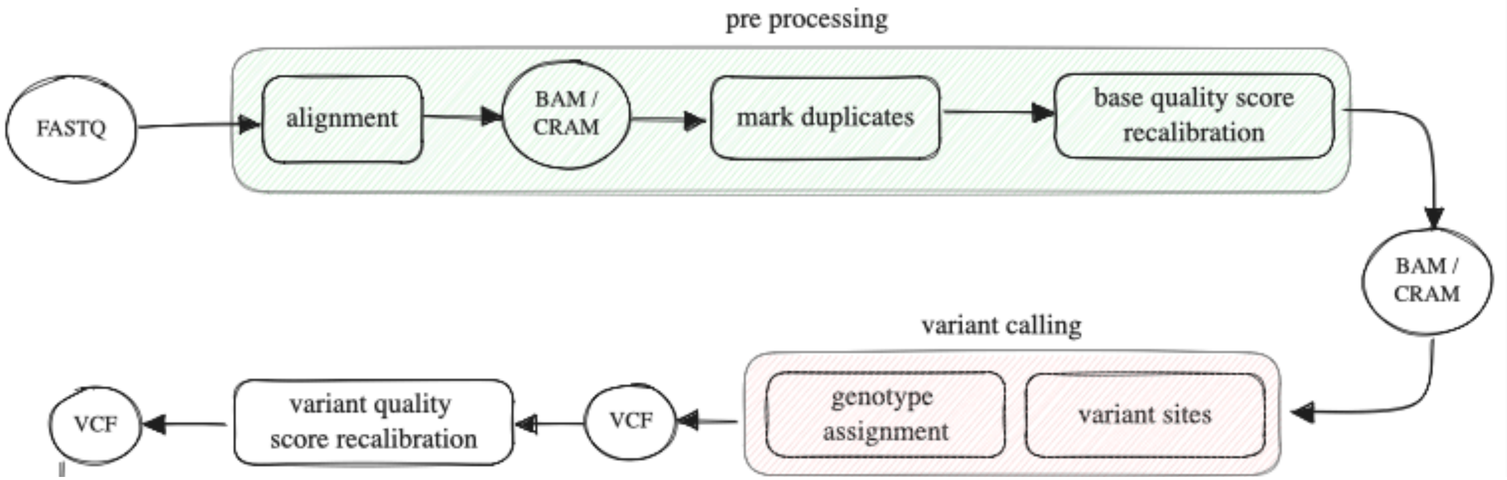
File format summary



Data Analysis Workflow

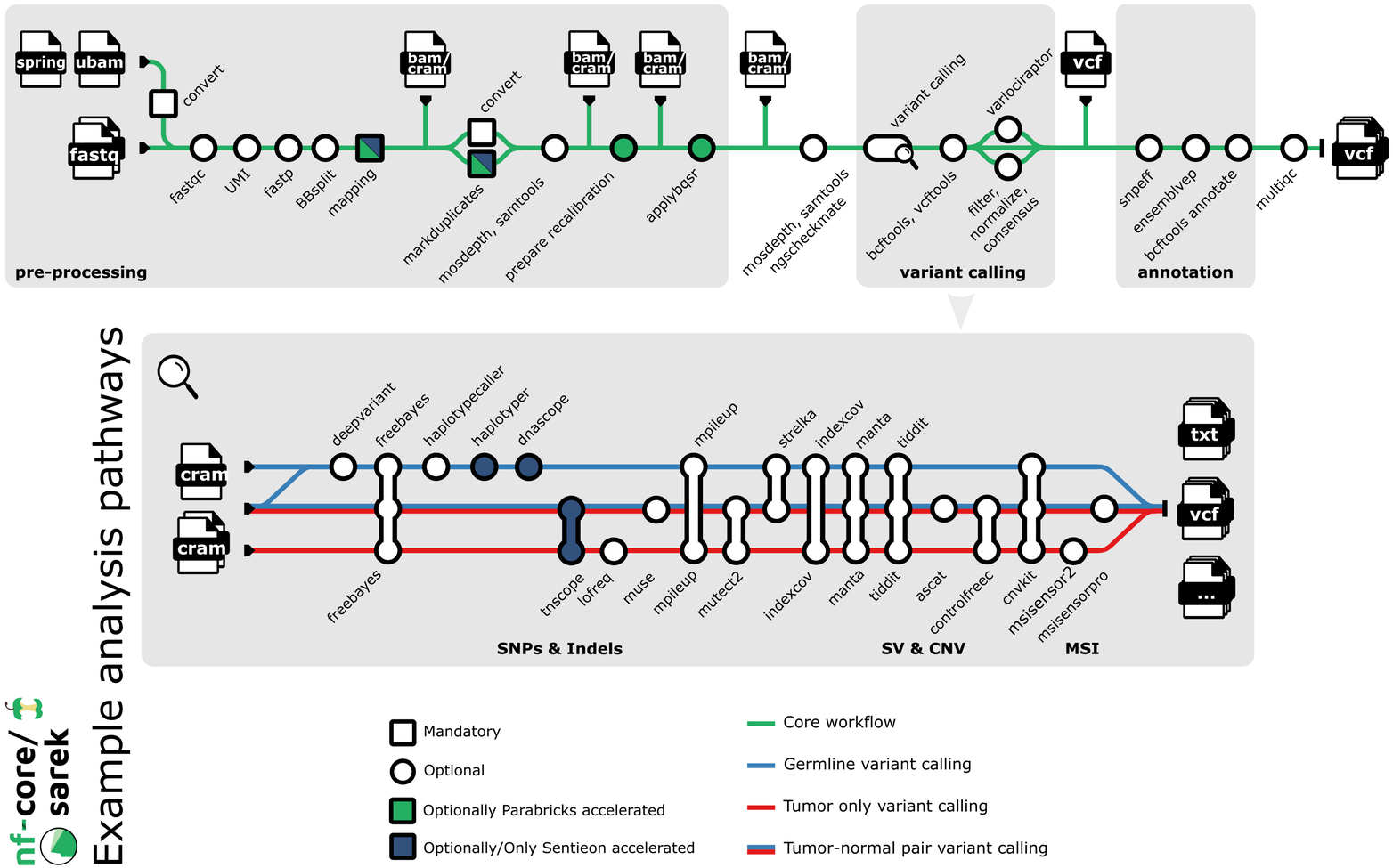
From reads to variant calls - GATK standards

1. Process raw sequencing data
2. Align pre-processed reads to the reference genome
3. Post-process aligned reads
4. Identify variants of interest.



nf-core/sarek

https://nf-co.re/sarek/

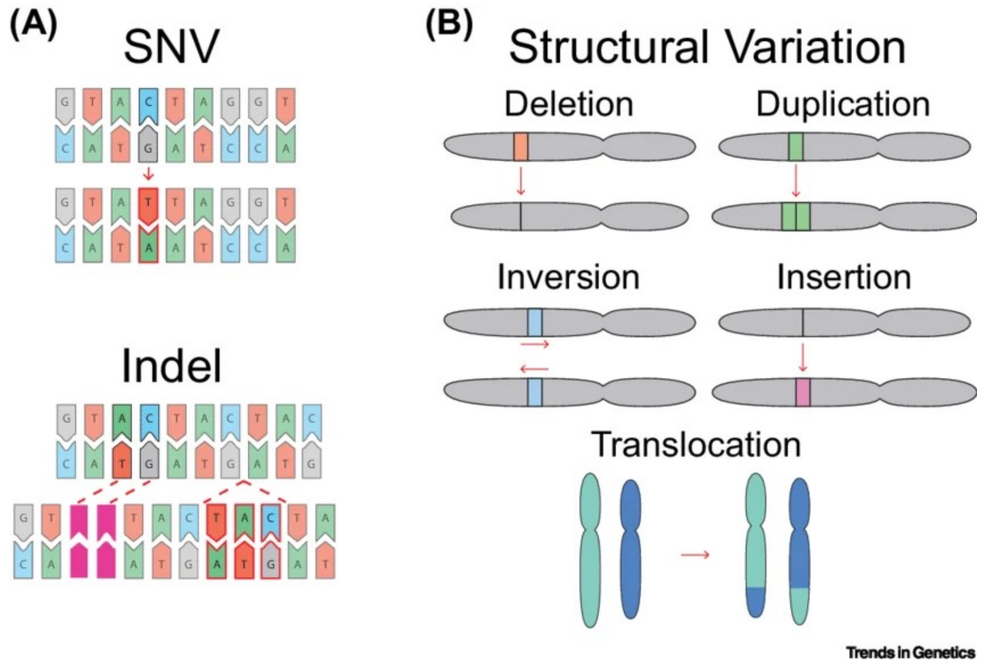


Adapted from: Fellows Yates, James A., et al. PeerJ 9 (2021).

What’s next?

Variant Calling

Variant calling aims at identifying single nucleotide variants (SNVs), small insertions and deletion (INDELS), copy number alterations (CNA) and structural variants (SV) from next generation sequencing data.



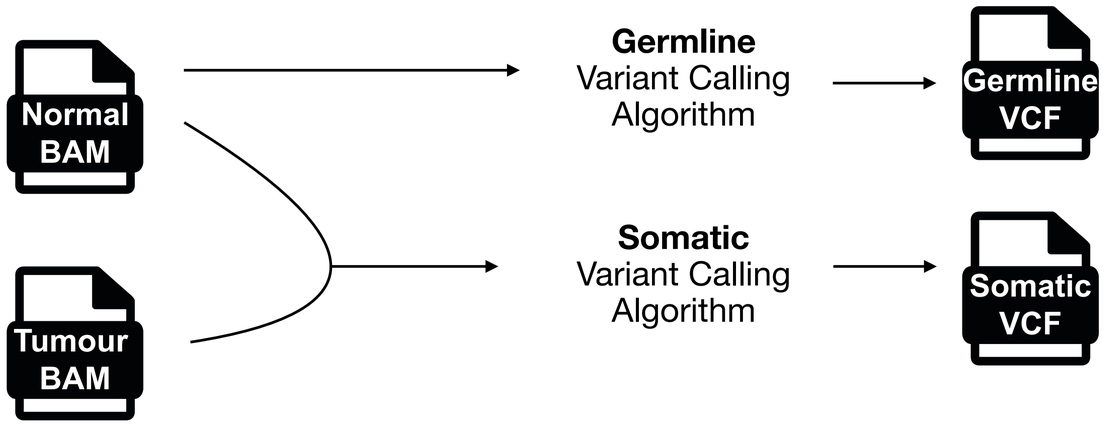
Nesta, A. V., Tafur, D., & Beck, C. R. (2021). Hotspots of human mutation. *Trends in Genetics*, *37*(8), 717-729.

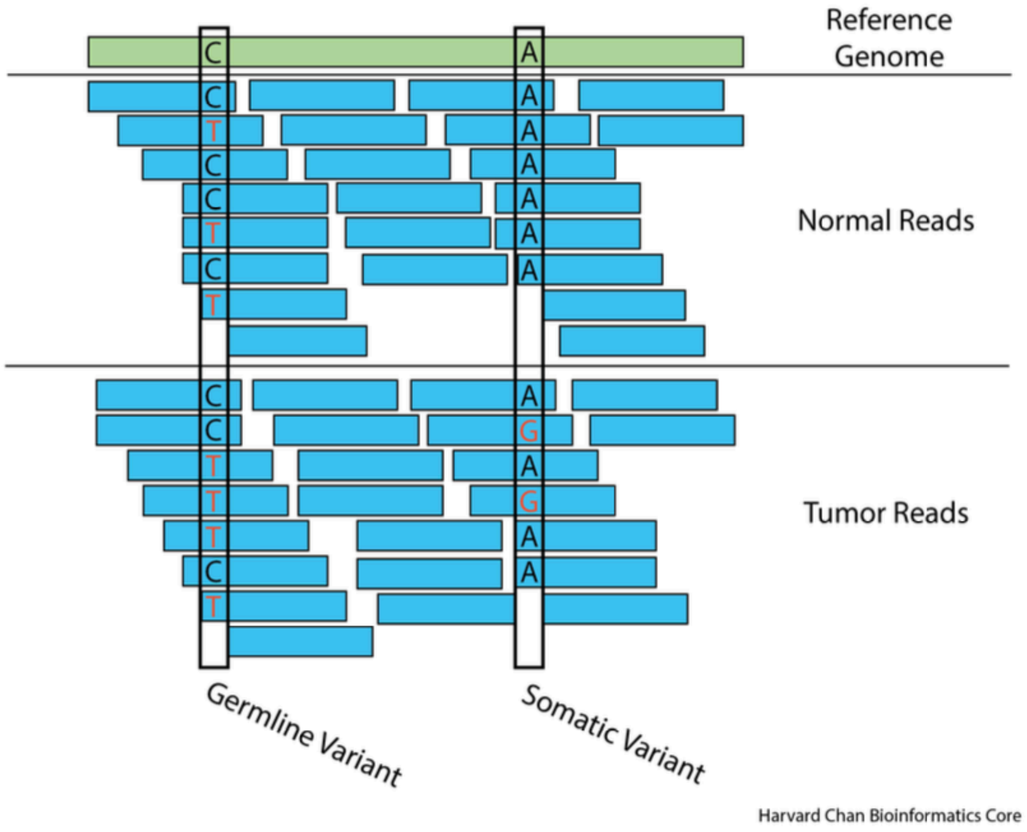
Germline vs Somatic

Small Variant Calling

Germline mutations are changes to your DNA that you inherit from the egg and sperm cells and it is passes directly from a parent to a child.
All the cells of the body inherit germline mutations.

Somatic mutations are changes to DNA that happens after conception, are not present in every cell in the body, and are not passed from parent to child.
Somatic mutations represent the major causes of cancer.

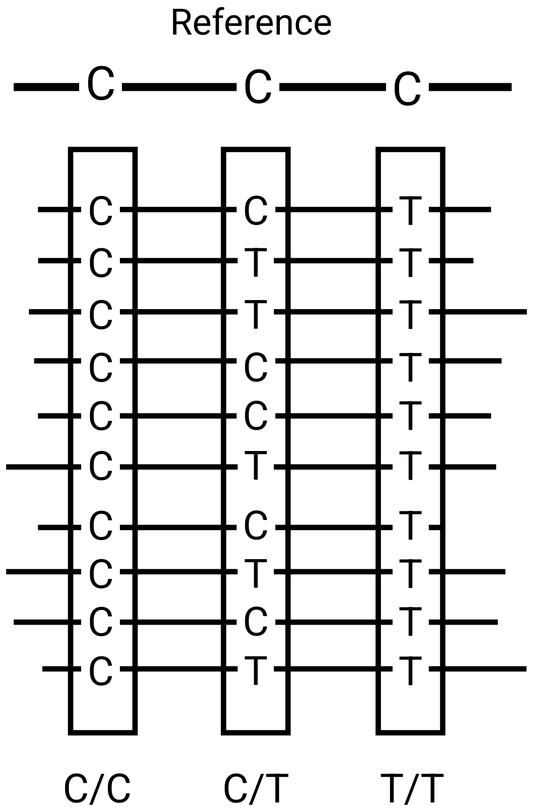


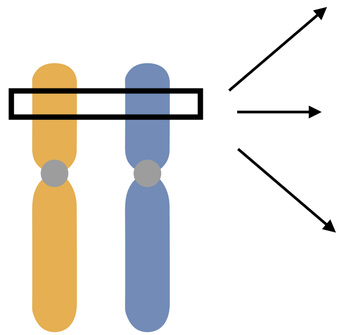


How variant calling works

Small Variant Calling

Genotypes: the specific combination of alleles (gene variants) inherited from its parents at a particular genomic locus





	Reference: C		REF/ALT	Depth DP	Allelic Depth AD	Allele Frequency AF
C/C	Homozygous	wild-type	0/0	20	10,10	0
C/T	Heterozygous		0/1	20	10,10	0.5
T/T	Homozygous	mutated	1/1	20	10,10	1

AF = N REF / (N REF + N ALT) = N REF / DP

VCF files

How a variant call is recorded



VCF header

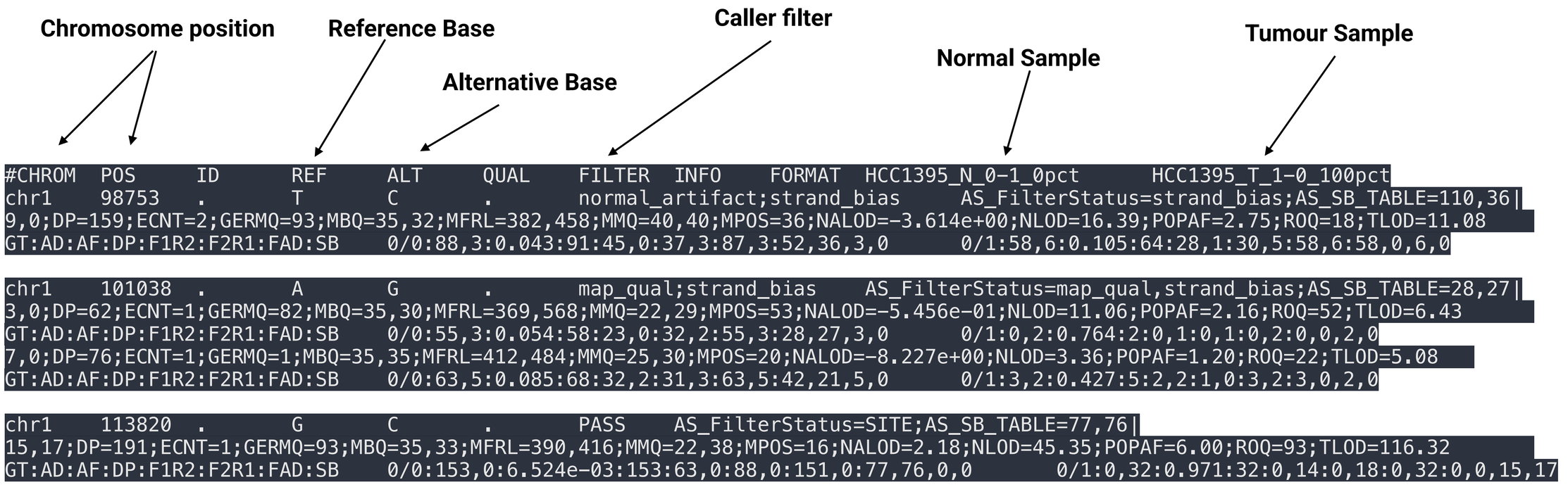
```
##fileformat=VCFv4.2
##FILTER=<ID=FAIL,Description="Fail the site if all alleles fail but for different reasons.">
##FILTER=<ID=PASS,Description="Site contains at least one allele that passes filters">
##FORMAT=<ID=AD,Number=R,Type=Integer,Description="Allelic depths for the ref and alt alleles in the order listed">
##FORMAT=<ID=AF,Number=A,Type=Float,Description="Allele fractions of alternate alleles in the tumor">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Approximate read depth (reads with MQ=255 or with bad mates are filtered)">
##FORMAT=<ID=F1R2,Number=R,Type=Integer,Description="Count of reads in F1R2 pair orientation supporting each allele">
##FORMAT=<ID=F2R1,Number=R,Type=Integer,Description="Count of reads in F2R1 pair orientation supporting each allele">
##FORMAT=<ID=FAD,Number=R,Type=Integer,Description="Count of fragments supporting each allele.">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=PGT,Number=1,Type=String,Description="Physical phasing haplotype information, describing how the alternate alleles
are phased in relation to one another; will always be heterozygous and is not intended to describe called alleles”>
```

VCF body

```
#CHROM  POS     ID      REF     ALT     QUAL    FILTER  INFO    FORMAT  HCC1395_N_0-1_0pct      HCC1395_T_1-4_20pct
chr1    14699   .       C       G       .       normal_artifact;strand_bias     AS_FilterStatus=strand_bias;AS_SB_TABLE=44,43|
0,10;DP=101;ECNT=1;GERMQ=77;MBQ=27,35;MFRL=382,460;MMQ=25,30;MPOS=49;NALOD=-3.116e+00;NLOD=12.87;POPAF=0.428;ROQ=89;TLOD=18.65
GT:AD:AF:DP:F1R2:F2R1:FAD:SB    0/0:74,3:0.051:77:35,1:34,2:73,3:40,34,0,3      0/1:13,7:0.400:20:5,3:6,4:11,7:4,9,0,7
chr1    16977   .       G       A       .       map_qual        AS_FilterStatus=map_qual;AS_SB_TABLE=74,48|
4,7;DP=134;ECNT=1;GERMQ=13;MBQ=35,34;MFRL=365,593;MMQ=23,22;MPOS=38;NALOD=-1.757e+01;NLOD=7.52;POPAF=0.400;ROQ=67;TLOD=8.28
GT:AD:AF:DP:F1R2:F2R1:FAD:SB    0/0:120,7:0.062:127:49,4:66,3:120,7:72,48,0,7   0/1:2,4:0.624:6:2,2:0,2:2,4:2,0,4,0
```

VCF files

How a variant call is recorded



Format of genotype field

GT = genotype
AD = allele depth
AF = allele fraction of ALT
DP = read depth

Genotype information of normal sample

GT = 0/0
AD = 153,0
AF = 6.524e-03
DP = 153

Genotype information of tumour sample

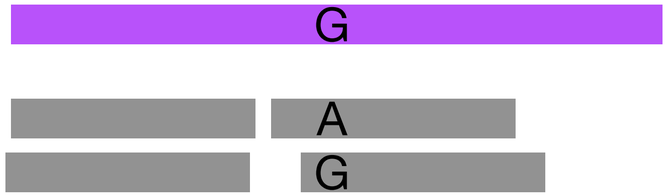
GT = 0/1
AD = 0,32
AF = 0.971
DP = 32

→ Reads covering only the alternative base (C)
Lower coverage

The importance of coverage

Lower coverage increases biases

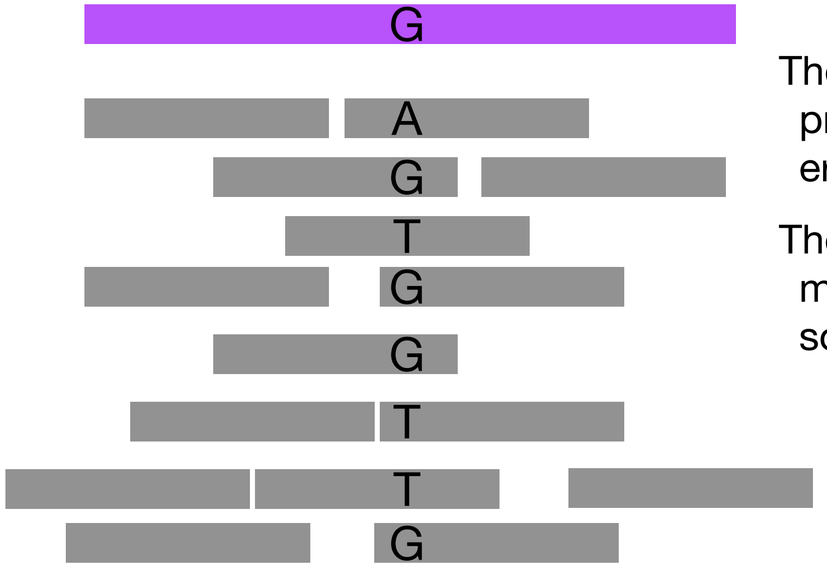
Low Coverage (2x)



There is a mutation G>A from the reference is this:

1. True somatic variant
2. Sequencing Error

High Coverage (8x)



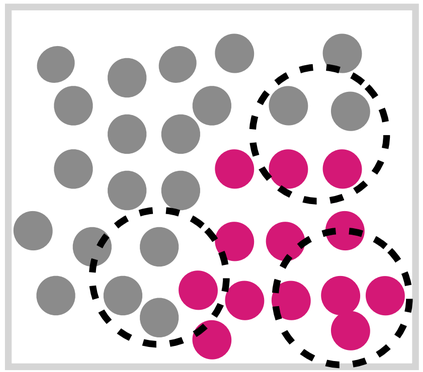
The mutation G>A is most probably a sequencing error

The mutation G>T is most probably a true somatic variant

The importance of purity

Sample purity is a bias in variant identification

Purity is the proportion of tumour cells in the sampled tissue



Sample A Purity = $\frac{2}{4} = 0.5$

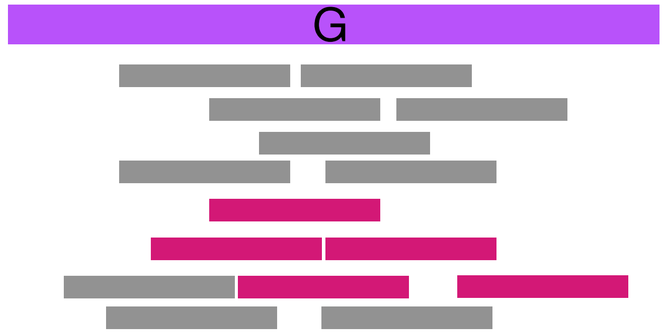
Sample B Purity = $\frac{4}{4} = 1$

Sample C Purity = $\frac{1}{4} = 0.25$

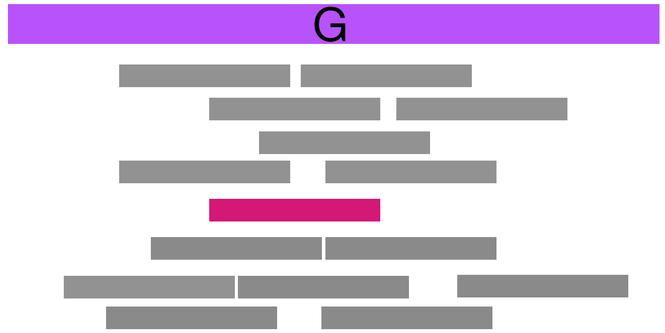
Normal tissue cell

Tumour tissue cell

High Purity



Low Purity



VCF filter

Each variant caller has its own filtering procedure

Strelka variant filters



```
chr21   5293968 .       C       T       .       LowEVS  SOMATIC;QSS=3;TQSS=1;NT=ref;QSS_NT=3;TQSS_NT=1;SGT=CC-
>CT;DP=264;MQ=47.27;MQ0=11;ReadPosRankSum=4.31;SNVSB=1.55;SomaticEVS=0.05 DP:FDP:SDP:SUBDP:AU:CU:
GU:TU   71:30:0:0:0,0:37,73:0,0:4,13    150:58:0:0:0,0:80,130:0,0:12,42
```



```
chr21   5294232 .       C       T       .       PASS    SOMATIC;QSS=85;TQSS=2;NT=ref;QSS_NT=85;TQSS_NT=2;SGT=CC-
>CT;DP=359;MQ=33.60;MQ0=108;ReadPosRankSum=-0.20;SNVSB=0.00;SomaticEVS=9.05     DP:FDP:SDP:SUBDP:AU:CU:GU:TU
40:5:0:0:0,0:35,115:0,0:0,3     82:9:0:0:0,0:45,201:0,0:28,35
```



```
chr21   5294838 .       C       T       .       LowEVS  SOMATIC;QSS=23;TQSS=2;NT=het;QSS_NT=23;TQSS_NT=2;SGT=CT-
>CC;DP=575;MQ=28.21;MQ0=222;ReadPosRankSum=-2.24;SNVSB=0.00;SomaticEVS=0.00     DP:FDP:SDP:SUBDP:AU:CU:GU:TU
57:26:0:0:0,0:19,87:0,0:12,53   129:42:0:0:0,1:74,259:0,1:13,92
```

Strelka available filters:

- PASS: mutation is a true somatic variant
- Low somatic Empirical Variant Score (SomaticEVS) is below threshold
- Low depth: Tumor or normal sample read depth at this locus is below 2

VCF filter

Each variant caller has its own filtering procedure



Mutect2 variant filters



```
chr21   5294027 .       G       C       .       map_qual;normal_artifact;strand_bias;weak_evidence
AS_FilterStatus=weak_evidence,map_qual,strand_bias;AS_SB_TABLE=125,117|0,7;DP=252;ECNT=1;GERMQ=93;MBQ=3
4,34;MFRL=392,335;MMQ=60,24;MPOS=30;NALOD=-1.213e+01;NLOD=13.62;POPAF=6.00;ROQ=30;TLOD=3.01     GT:AD:AF:DP:F1R2:F2R1:FAD:SB
0/0:117,5:0.049:122:66,3:43,2:115,5:53,64,0,5   0/1:125,2:0.024:127:59,1:56,1:1
23,2:72,53,0,2
```



```
chr21   5294232 .       C       T       .       PASS    AS_FilterStatus=SITE;AS_SB_TABLE=105,102|
24,30;DP=274;ECNT=1;GERMQ=93;MBQ=33,34;MFRL=379,366;MMQ=60,60;MPOS=40;NALOD=2.05;NLOD=32.50;POPAF=6.00;ROQ=93;TLOD=161.79
GT:AD:AF:DP:F1R2:F2R1:FAD:SB    0/0:112,0:8.860e-03:112:59,0:47,0:108,0:58,54,0,0
0/1:95,54:0.374:149:48,23:40,30:91,54:47,48,24,30
```



```
chr21   5298191 .       C       G       .       haplotype;normal_artifact       AS_FilterStatus=SITE;AS_SB_TABLE=130,130|
13,15;DP=288;ECNT=2;GERMQ=93;MBQ=33,20;MFRL=355,355;MMQ=60,60;MPOS=9;NALOD=1.88;NLOD=43.72;POPAF=6.00;ROQ=21;TLOD=5.10
GT:AD:AF:DP:F1R2:F2R1:FAD:PGT:PID:PS:SB 0|0:140,15:6.683e-03:155:63,0:51,0:151,9:0|1:5298191_C_G:5298191:66,74,7,8      0|
1:120,13:0.016:133:43,0:47,1:128,8:0|1:5298191_C_G:5298191:64,56,6,7
```

Mutect2 available filters:

- PASS: mutation is a true somatic variant
- Mapping quality: ref - alt median mapping quality
- Base quality: alt median base quality
- Clustered events: lot of close mutations in the tumour
- Normal artefacts
- Haplotype: variant near filtered variant on same haplotype
- Germline: Evidence indicates this site is germline, not somatic
- ……

BED files

Variant Calling



- BED (Browser Extensible Data) format provides a flexible way to define the data lines that are displayed in an annotation track.
- BED lines have three required fields and nine additional optional fields.
- BED identifies a region you want to inspect or analyze.

The first three required BED fields are:

1. **chrom** - The name of the chromosome (e.g. chr3, chrY, chr2_random) or scaffold (e.g. scaffold10671).

2. **chromStart** - The starting position of the feature in the chromosome or scaffold. The first base in a chromosome is numbered 0.

4. **chromEnd** - The ending position of the feature in the chromosome or scaffold. The *chromEnd* base is not included in the display of the feature, however, the number in **position format** will be represented. For example, the first 100 bases of chromosome 1 are defined as *chrom=1, chromStart=0, chromEnd=100*, and span the bases numbered 0-99 in our software (not 0-100), but will represent the position notation chr1:1-100.

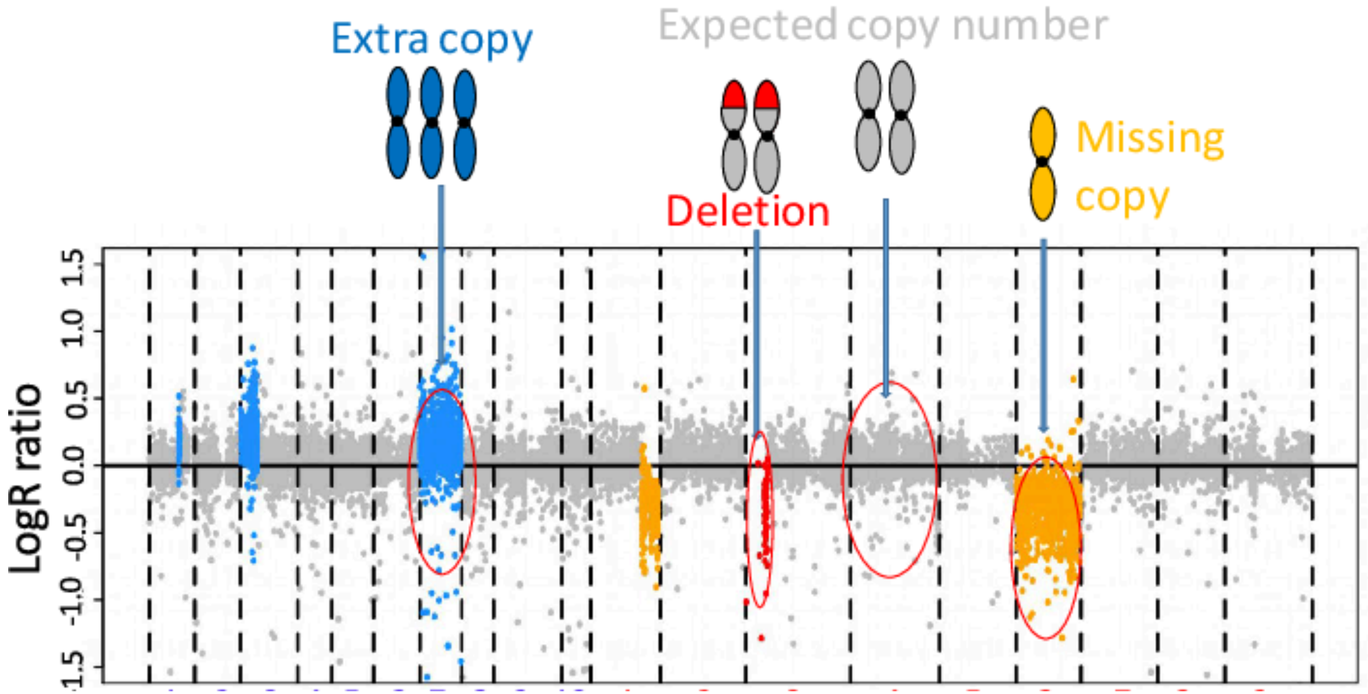
Copy Number calling

Variant Calling

Principles

- More DNA copies → more reads aligning to that region

- Read depth (coverage) is normalized against a matched normal and $\log_2$-transformed: $\log_2$(tumor/normal)



Katche, Elizabeth Ihien, et al. "Genetic factors inherited from both diploid parents interact to affect genome stability and fertility in resynthesized allotetraploid Brassica napus." *G3: Genes, Genomes, Genetics* 13.8 (2023):

Copy Number calling

File format

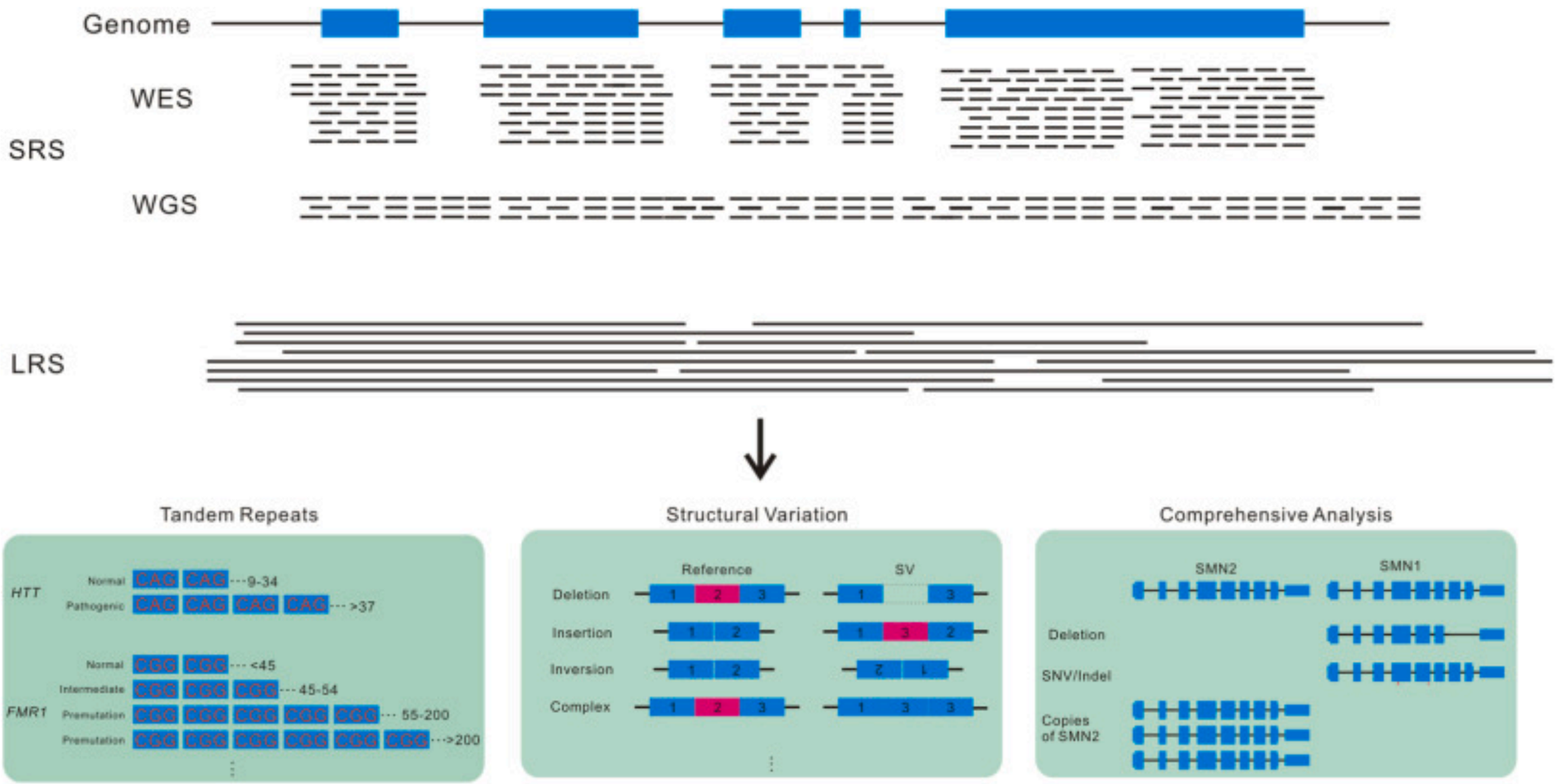
chromosome	start	end	cn
chr21	5009996	5045716	1
chr21	5045792	5053240	5
chr21	5054380	5108720	1
chr21	5108720	5112900	3
chr21	5112900	5159792	1
chr21	5159792	5160628	5
chr21	5161540	5165568	1
chr21	5165720	5511976	4
chr21	5512052	5519576	3

Short-read vs long-read sequencing

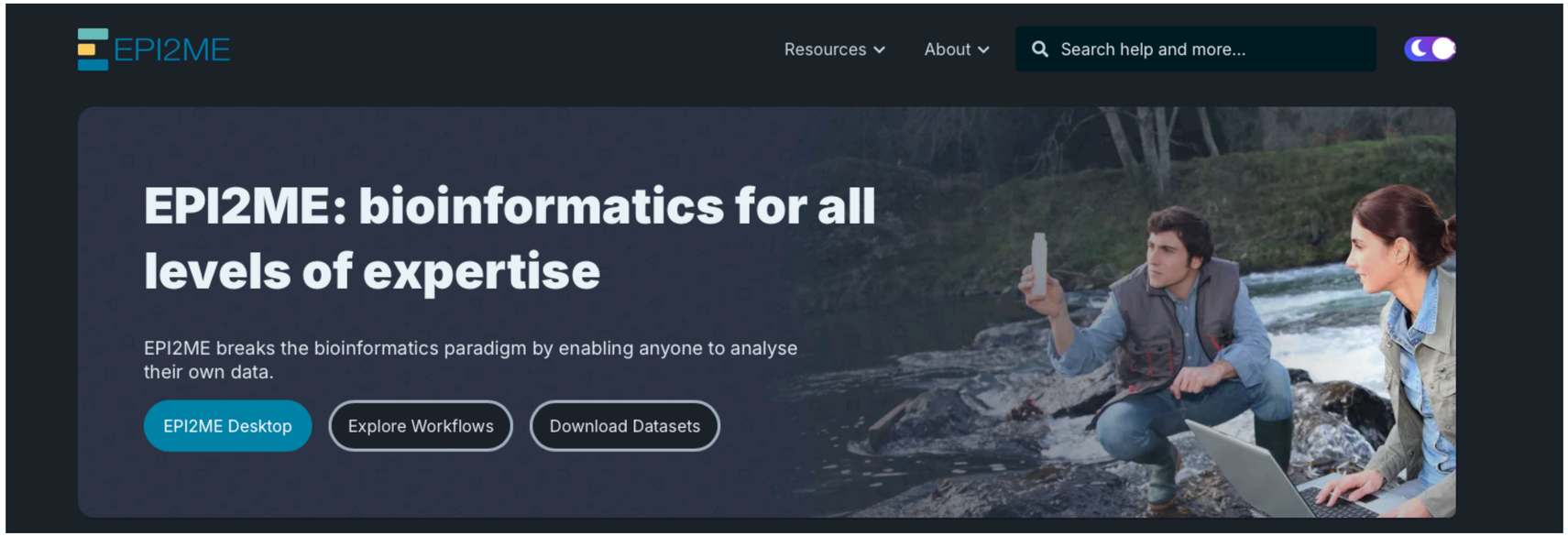
The choice of sequencing technology shapes what biological questions you can answer.

	Short-read (Illumina)	Long-read (PacBio / ONT)
Read length	150–300 bp	1 kb – 100+ kb
Accuracy	Very high (Q30+)	Lower raw accuracy, but improving rapidly
Throughput	Very high	Lower, though increasing with newer platforms
Cost per Gb	Low	Higher, dropping fast
SNV / indel calling	Excellent	Good
SVs & CNVs	Limited - short reads struggle to span breakpoints	Excellent - reads cross the entire variant
Repetitive regions	Poor - reads too short to map uniquely	Excellent - reads span through repeats
Methylation	Requires bisulfite conversion (extra step)	Direct detection, no conversion needed
Typical use cases	WGS, WES, gene panels, somatic calling	SV detection, de novo assembly, phasing, methylation

Long read Advantages



How to process long-reads data



Basecalling

Dorado/Guppy

wf-basecalling

Alignment

minimap2

wf-aligment

Variant Calling

Clair3/ClairS

wf-humanvariation

Methylation

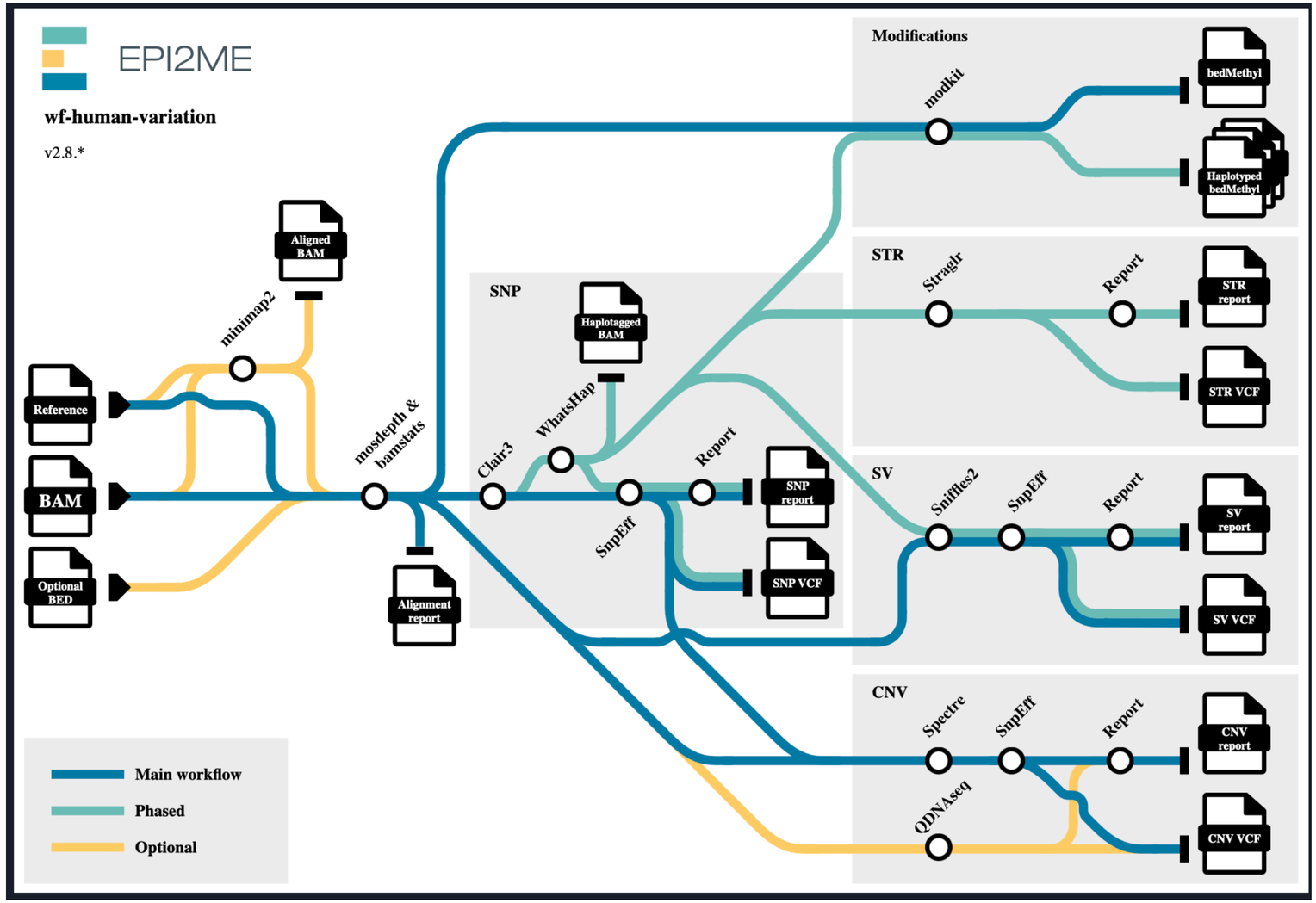
modkit

How to process long-reads data

wf-basecalling

↓

wf-aligment →

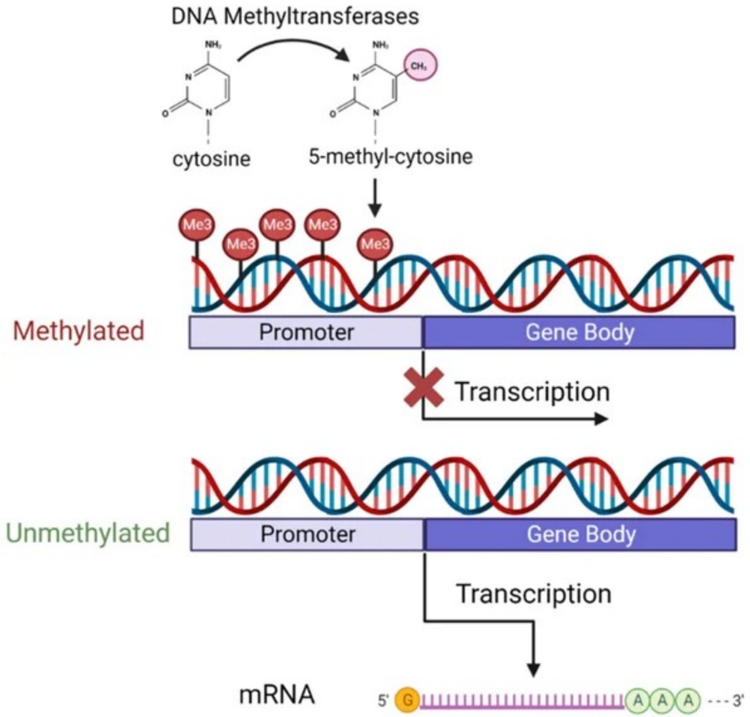


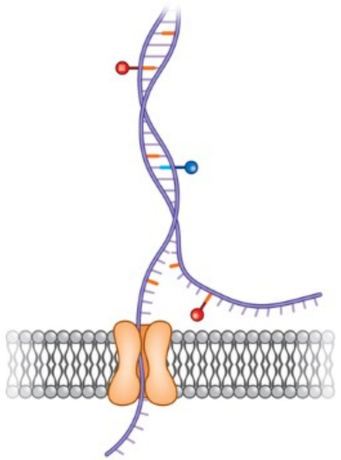

```
nextflow run epi2me-labs/wf-human-variation \
    --bam 'wf-human-variation-demo/demo.bam' \
    --ref 'wf-human-variation-demo/demo.fasta' \
    --bed 'wf-human-variation-demo/demo.bed' \
    --sample_name 'DEMO' \
    --snp \
    --sv \
    --mod \
    --phased \
    -profile standard
```

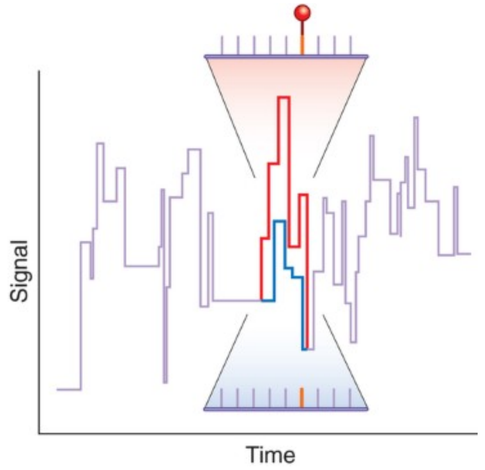

https://epi2me.nanoporetech.com/epi2me-docs/workflows/wf-human-variation/

Methylation

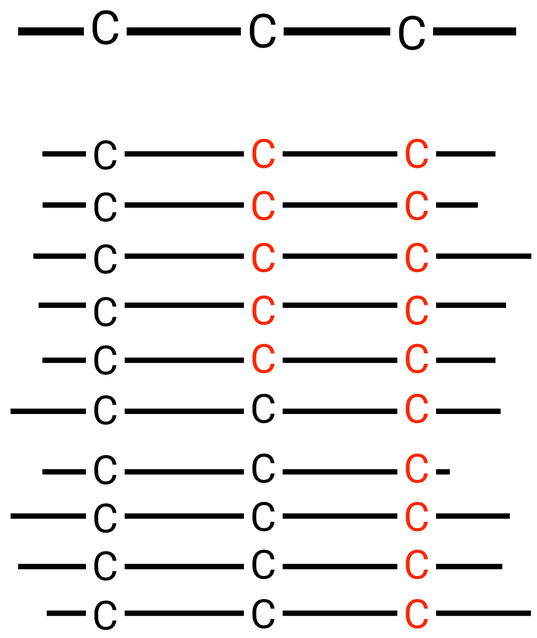
How to detect epigenetic modification?







Reference



β = N reads methylated / Tot Reads

0 0.5 1

Methylation

Modkit




```
modkit pileup “sample1.bam” “output_sample1_methylation.bed” \
  —ref “reference” \
  --ignore h \
  --cpg
```

%mod = (N_mod / N_valid_cov) *100

chrom	start	end	mod	score	strand	Nvalid_cov	%mod	N_mod	N_can	N_other	N_del	N_fail	N_diff
chr21	8990197	8990198	m	113	.	113	6.19	7	106	0	0	2	12
chr21	8990219	8990220	m	35	.	35	100.00	35	0	0	1	4	54
chr21	8990221	8990222	m	131	.	131	38.93	51	80	0	1	11	2
chr21	8990223	8990224	m	60	.	60	93.33	56	4	0	0	4	40
chr21	8990247	8990248	m	134	.	134	4.48	6	128	0	1	8	0
chr21	8990275	8990276	m	120	.	120	0.00	0	120	0	1	16	3
chr21	8990282	8990283	m	137	.	137	0.00	0	137	0	1	4	1
chr21	8990286	8990287	m	98	.	98	3.06	3	95	0	1	6	21
chr21	8990334	8990335	m	130	.	130	2.31	3	127	0	3	5	0
chr21	8990345	8990346	m	128	.	128	1.56	2	126	0	2	3	6
chr21	8990431	8990432	m	106	.	106	39.62	42	64	0	1	13	13
chr21	8990608	8990609	m	135	.	135	0.00	0	135	0	0	5	0
chr21	8990623	8990624	m	91	.	91	0.00	0	91	0	0	7	27

Methylation in IGV



References

Tool	Paper	GitHub	Link
Nextflow	Nextflow enables reproducible	https://github.com/nextflow-io/nextflow	https://www.nextflow.io
nf-core	The nf-core framework for community-	https://github.com/nf-core	https://nf-co.re
nf-core/sarek	Sarek: A portable workflow for	https://github.com/nf-core/sarek	https://nf-co.re/sarek
GATK Best Practices	The Genome Analysis Toolkit: a MapReduce	https://github.com/broadinstitute/gatk	https://gatk.broadinstitute.org
EPI2ME wf-basecalling	-	https://github.com/epi2me-labs/wf-basecalling	-
EPI2ME wf-human-variation	-	https://github.com/epi2me-labs/wf-human-variation	-
FastQC	-	https://github.com/s-andrews/FastQC	https://www.bioinformatics.babraham.ac.uk/projects/fastqc/
fastp	fastp: an ultra-fast all-in-one FASTQ	https://github.com/OpenGene/fastp	-
MultiQC	MultiQC: summarize analysis results	https://github.com/MultiQC/MultiQC	https://multiqc.info
Dorado	-	https://github.com/nanoporetech/dorado	-
BWA-MEM	Aligning sequence reads, clone	https://github.com/lh3/bwa	-
minimap2	Minimap2: pairwise alignment for	https://github.com/lh3/minimap2	-
SAMtools/bcftools	Twelve years of SAMtools and BCFtools	https://github.com/samtools/samtools https://github.com/samtools/bcftools	https://www.htslib.org
HaplotypeCaller	Scaling accurate genetic variant	https://github.com/broadinstitute/gatk	https://gatk.broadinstitute.org
DeepVariant	A universal SNP and small-indel variant	https://github.com/google/deepvariant	-
Mutect2	Calling somatic SNVs and indels with	https://github.com/broadinstitute/gatk	https://gatk.broadinstitute.org
Strelka2	Strelka2: fast and accurate calling of	https://github.com/Illumina/strelka	-
Clair3	Symphonizing pileup and full-alignment for	https://github.com/HKU-BAL/Clair3	-
IGV	Integrative genomics viewer	https://github.com/igvteam/igv	https://igv.org